Some notes on Statistical Learning Theory

Course of Machine Learning

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Statistical learning theory provides the theoretical foundation for understanding machine learning. It asks a fundamental question: given that we've chosen a hypothesis class $\mathcal{H}$ and collected a training set of some size n, what can we guarantee about the quality of the predictor we'll find? Put differently, under what conditions is learning actually possible?

This is not a question about specific algorithms or datasets, but rather about the general structure of the learning problem. The theory gives us precise mathematical answers that apply to any learning scenario, revealing universal principles that govern whether and how quickly we can learn.

Learning Algorithms and the Role of Hypothesis Classes

A learning algorithm $\mathcal{A}$ is fundamentally a procedure that takes training data and produces a predictor. However, a learning algorithm cannot be entirely free-form. It must operate within some structured framework, which is the **hypothesis class** $\mathcal{H}$. The hypothesis class specifies the set of candidate functions the algorithm is allowed to choose from. In other words, we're saying to the algorithm: "Choose the best predictor you can find, but it must be from this predefined set."

This constraint is what we call the algorithm's **inductive bias**. It's the learner's prior belief about what kind of solution is likely to work. Every learning algorithm must encode this bias somehow, whether explicitly or implicitly. The choice of hypothesis class is therefore one of the most consequential decisions in machine learning practice.

Formally, we can denote the predictor returned by algorithm $\mathcal{A}$ when applied to training set $\mathcal{T}$ as $h_{\mathcal{T}}^{\mathcal{A}}$. This notation emphasizes that the learned predictor depends both on which algorithm we use and which training data we received.

Empirical Risk Minimization

Among all possible learning algorithms, one stands out for its simplicity and prevalence: **Empirical Risk Minimization** (ERM). The ERM algorithm is beautifully straightforward: given a training set $\mathcal{T}$ and a hypothesis class $\mathcal{H}$, it returns the predictor in $\mathcal{H}$ that achieves the lowest training error:

$$h_{\mathcal{T}} = \operatorname*{argmin}_{h \in \mathcal{H}} \overline{\mathcal{R}}_{\mathcal{T}}(h)$$

where $\overline{\mathcal{R}}_{\mathcal{T}}(h)$ is the empirical risk—the average loss on the training set.

The intuition behind ERM is compelling: if we want a predictor that works well in general, we should choose one that works well on the data we have. This greedy approach seems natural, almost obvious. Yet this apparent simplicity masks a subtle danger. The algorithm is optimizing with respect to the training data, and it's entirely possible for the training data to be misleading. A predictor could fit the training set perfectly while generalizing poorly to the rest of the population—the problem of overfitting that we've discussed before.

The central question of statistical learning theory is therefore: under what conditions is ERM (or other learning algorithms) actually reliable? When does minimizing training error lead to low true error? The answer depends critically on the hypothesis class.

The Fundamental Question

The core question that motivates statistical learning theory is the following: given a hypothesis class $\mathcal{H}$, a loss function L, and a distribution p over the data, can we guarantee that applying a learning algorithm like ERM to a sufficiently large training set will produce a predictor with small true risk?

This question is not about specific data or specific problems. Rather, it asks: what properties of the hypothesis class determine whether learning is possible at all? Some hypothesis classes are "learnable" - meaning that with enough data, we can find good predictors. Others are "too complex" and resist learning because the set of possible functions is so large that we can always find functions fitting the training data perfectly while failing on new data.

The theory answers this question, showing that learnability depends on:

1. The size of the hypothesis class (roughly, how many functions it contains)
2. The amount of training data available
3. How accurately and confidently we need to learn (our tolerance for error)

The relationships between these quantities are not obvious, understanding them is the essence of statistical learning theory.

Learning with Finite Hypothesis Classes Under Realizability

The simplest type of restriction on $\mathcal{H}$ is assuming the number of possible predictors is upper bounded. In this case, choosing $h_{\mathcal{T}}$ will not overfit, provided $\mathcal{T}$ is sufficiently large, that is a sufficient number of examples of the application of f is available[1].

Let us first limit ourselves to the case when there exists some predictor in $h^* \in \mathcal{H}$ which does not make any error in classifying items in $\mathcal{X}$, that is such that

$$\mathcal{R}_{p_M,f}(h^*) = \mathop{\mathbb{E}}_{p_M,f}[\, L(h^*(\mathbf{x}), f(\mathbf{x}))\,] = \mathop{\mathbb{E}}_{p_M,f}[\, |\mathbf{x} \in \mathcal{X} \,:\, h^*(\mathbf{x}) \neq f(\mathbf{x})|\,] = 0$$

This is denoted as **realizability assumption** and, since the 0-1- loss is a non negative function, it implies that[2] it correctly classifies all the elements in any subset of $\mathcal{X}$, that is for any possible set $\mathbf{X}$ of elements sampled from $\mathcal{X}$, which implies that for any training set $\mathcal{T} = (\mathbf{X}, \mathbf{t})$, where $t_i = f(\mathbf{x}_i)$ by assumption, it results

[1] Our considerations will be limited here to the task of binary classification with $0-1$ loss function, and we will refer to the simpler scenario where there is a functional, albeit unknown, relation between items and target values, i.e. classes.

[2] With probability 1, since the expectation equal to 0 does not rule out the possibility of single points (that is sets of measure 0) in $\mathcal{X}$ misclassified by h^*.

$$\overline{\mathcal{R}}_{\mathcal{T}}(h^*) = \frac{1}{|\mathcal{T}|} \sum_{(x,t)\in\mathcal{T}} L(h^*(x), t) = \frac{|(\mathbf{x}, t) \in \mathcal{T} \ : \ h^*(\mathbf{x}) \neq t|}{|\mathcal{T}|} = 0$$

Realizability is a strong assumption—it assumes our hypothesis class contains a perfect model of the world. While unrealistic in practice, this idealized setting isolates the challenge of learning from finite samples, separate from the problem of approximating a complex reality with a restricted model class.

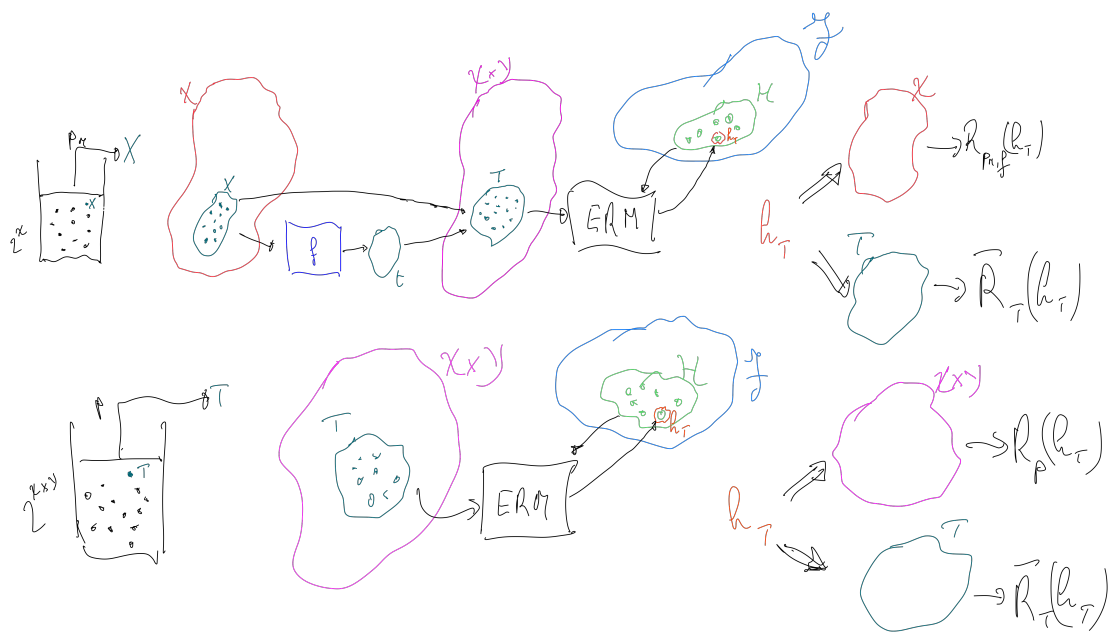

Figura 1. Sketch of the situation. Above, the case of a deterministic relation between item and target value. We assume a set $\mathbf{X}$ of items has been sampled from the population $\mathcal{X}$ under probability distribution $p_M(\mathbf{x})$ and the corresponding target values $\mathbf{t}$ are associated by applying the (unknown) function f. ERM (indeed, any learning algorithm) selects, given the resulting training set $\mathcal{T} = (\mathbf{X}, \mathbf{t})$, a predictor $h_{\mathcal{T}}$ from the hypothesis class $\mathcal{H}$, which is a subset of the set $\mathcal{F}$ of all possible predictors. When applied to $\mathcal{T}$, the selected predictor has minimum empirical risk $\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}})$, while when the expectation over all possible training sets, sampled selecting $\mathbf{X}$ according to p_M, is taken into account, the risk $\mathcal{R}_{p_M,f}(h_{\mathcal{T}})$ is not necessarily the minimum one: this means that $h_{\mathcal{T}}$ behaves well on known data and more poorly when applied to new data (overfitting). Observe that under the realizability hypothesis, $\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) = 0$. Below, the same is shown in the more general case when a probabilistic relation is assumed between items and target values. Here the same training set $\mathcal{T}$ is assumed sampled from the population $\mathcal{X} \times \mathcal{Y}$ under distribution $p(\mathbf{x}, t) = p_M(\mathbf{x})p_c(t|\mathbf{x})$.

When we apply Empirical Risk Minimization (ERM) to a training set, the algorithm finds a predictor with minimum training error.

Realizability also implies that for any training set $\mathcal{T}$ the predictor returned by ERM is optimal when applied to $\mathcal{T}$, that is $\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) = 0$: this derives from the observation that, given $\mathcal{T}$, there exists at least one predictor (that is h^*) which correctly classifies all elements in it. Since ERM returns a predictor, say it $h_{\mathcal{T}}$, which best classifies such elements it must be $\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) = 0$. Note that, as a special case, it might be $h_{\mathcal{T}} = h^*$, that is ERM returns precisely h^*: in this case, the predictor returned by ERM on $\mathcal{T}$ is optimal on the whole elements population $\mathcal{X}$. However, it may also happen that $h_{\mathcal{T}} \neq h^*$, which implies that the predictor returned by ERM on $\mathcal{T}$ is optimal on $\mathcal{T}$ (that is $\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) = 0$), but it may return incorrect predictions on some elements not in $\mathcal{T}$ (that is, $\mathcal{R}_{p_M,f}(h_{\mathcal{T}}) > 0$).

We are then guaranteed that

$$\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) = 0$$

This is encouraging: ERM returns a predictor that perfectly fits the training data. However, this is both weaker and more subtle than it first appears. The predictor $h_{\mathcal{T}}$ returned by ERM might not be h^*. In fact, ERM might return any predictor from the set of all functions that achieve zero training error—a set that could contain many functions. Some of these functions are genuinely good (close in behavior to h^*), while others are misleadingly good: they happened to classify all training examples correctly purely by coincidence, despite being terrible on the rest of the population. This is the core overfitting problem.

The challenge is therefore to understand when we can trust that zero training error implies low true error. This depends on whether the training set is *representative* of the population.

Bad Predictors and Datasets

To formalize the notion of a misleading training set, we introduce a parameter $\varepsilon > 0$ representing our acceptable error tolerance. We call a predictor $h \in \mathcal{H}$ **bad** if its true risk exceeds this threshold:

$$\mathcal{R}_{p_M,f}(h) > \varepsilon$$

A bad predictor is one that makes mistakes on more than a fraction ε of the population—too many mistakes to be practically useful. Let $\mathcal{H}_B$ denote the set of all bad predictors.

A training set $\mathcal{T}$ is called **bad** (or misleading) if at least one bad predictor achieves zero training error on it:

$$\exists h \in \mathcal{H}_B \,:\, \overline{\mathcal{R}}_{\mathcal{T}}(h) = 0$$

This is the failure case we want to avoid. A bad training set creates the illusion that a poor predictor is good, because the predictor happens to classify all training examples correctly despite making many errors elsewhere. If ERM returns such a bad predictor, we call the training set *very bad*. So, a training set is very bad if applying ERM on it returns a predictor which performs no error in classifying its elements, while failing in classifying at least a

fraction ε of all the possible elements. The probability of obtaining a bad (or very bad) training set is what we need to bound.

Assume now that, given $\mathcal{H}$, we want to study for which values of $|\mathcal{T}| = n$ the probability that our training set (assuming that it results from independently sampling each element from $\mathcal{X}$ with distribution p_M and setting $t_i = f(\mathbf{x}_i)$ or by independently sampling from $\mathcal{X} \times \mathcal{Y}$ with distribution p) is a bad one is sufficiently small, for example, less than a given $\delta \in (0, 1)$. Hence,

$$\mathbb{P}_{\mathcal{T} \sim p^n} \left[\exists h \in \mathcal{H}_B \,:\, \overline{\mathcal{R}}_{\mathcal{T}}(h) = 0 \right] \leq \delta$$

Consider any single bad predictor $h \in \mathcal{H}_B$. By definition, h misclassifies at least a fraction ε of the population. When we draw a training set by sampling n objects independently from p_M, what is the probability that h happens to classify all n objects correctly?

Since h misclassifies at least a fraction ε of the population, it correctly classifies at most a fraction $1 - \varepsilon$. For a single random object, the probability of correct classification is at most $1 - \varepsilon$. For n independent objects, the probability that h correctly classifies all of them is at most $(1 - \varepsilon)^n$.

Using the inequality $1 - x \leq e^{-x}$ for $x \in [0, 1]$:

$$(1 - \varepsilon)^n \leq e^{-\varepsilon n}$$

This bounds the probability that a specific bad predictor fools us. But we are concerned about *any* bad predictor succeeding, not just one. The **union bound** states that the probability of at least one event occurring is bounded by the sum of individual event probabilities. Applying this:

$$\mathbb{P}_{\mathcal{T} \sim p^n} \left[\exists h \in \mathcal{H}_B \,:\, \overline{\mathcal{R}}_{\mathcal{T}}(h) = 0 \right] \leq \sum_{h \in \mathcal{H}_B} e^{-\varepsilon n} = |\mathcal{H}_B| \cdot e^{-\varepsilon n}$$

In the worst case, all predictors are bad, so $|\mathcal{H}_B| \leq |\mathcal{H}|$:

$$\mathbb{P}_{\mathcal{T} \sim p^n} \left[\exists h \in \mathcal{H}_B \,:\, \overline{\mathcal{R}}_{\mathcal{T}}(h) = 0 \right] \leq |\mathcal{H}| \cdot e^{-\varepsilon n}$$

So, we have shown that the probability that a random training set of size n is a bad one is at most $|\mathcal{H}| \cdot e^{-\varepsilon n}$.

Then setting

$$|\mathcal{H}| \cdot e^{-\varepsilon n} \leq \delta$$

implies that the probability that a random training set $\mathcal{T}$ is bad is itself bounded by δ.

Rearranging the above inequality to solve for the required training set size:

$$e^{-\varepsilon n} \leq \frac{\delta}{|\mathcal{H}|}$$

Taking natural logarithms of both sides (noting that $\ln$ is monotone increasing):

$$-\varepsilon n \leq \ln \frac{\delta}{|\mathcal{H}|} = \ln \delta - \ln |\mathcal{H}|$$

Dividing by $-\varepsilon$ (and reversing the inequality):

$$n \geq \frac{1}{\varepsilon}(\ln |\mathcal{H}| - \ln \delta) = \frac{1}{\varepsilon} \ln \frac{|\mathcal{H}|}{\delta}$$

Since n must be an integer, we have:

$$n \geq \left\lceil \frac{1}{\varepsilon} \ln \frac{|\mathcal{H}|}{\delta} \right\rceil \tag{1}$$

This is the **sample complexity** for learning with finite hypothesis classes under realizability. It specifies the minimum training set size needed to ensure that with probability at least $1 - \delta$, ERM returns a predictor whose true risk is at most ε.

The formula in Equation 1 deserves careful interpretation, as each component reveals important principles about learning.

The Role of Accuracy. The accuracy parameter ε appears in the denominator, creating an inverse relationship. If we want to halve the acceptable error (going from ε to $\varepsilon/2$), the required training set size doubles. More generally, if we reduce ε by a factor of c, sample complexity increases by a factor of c. This linear (non-logarithmic) dependence means that achieving high accuracy is fundamentally expensive in terms of data requirements. This reflects a deep fact about learning: there's no escape from accuracy's cost.

For example, with $|\mathcal{H}| = 1000$ and $\delta = 0.05$, requiring $\varepsilon = 0.1$ error needs $n \geq \frac{1}{0.1} \ln(20000) \approx 100$ examples. Requiring $\varepsilon = 0.01$ error needs $n \geq \frac{1}{0.01} \ln(20000) \approx 1000$ examples—ten times as much data for ten times better accuracy.

The Role of Confidence. The confidence parameter δ appears inside a logarithm in the numerator. If we want higher confidence by reducing δ by a factor of c (say from $\delta = 0.1$ to $\delta' = 0.01$, a 10-fold reduction), the required sample size increases by $\frac{1}{\varepsilon} \ln c$. For typical values, this is a modest increase. Increasing from 90% to 99% confidence (reducing δ from 0.1 to 0.01) requires adding only $\frac{1}{\varepsilon} \ln 10 \approx \frac{2.3}{\varepsilon}$ additional examples.

This logarithmic sensitivity to δ versus linear sensitivity to ε reveals an asymmetry: we can achieve high confidence relatively cheaply by collecting more data, but achieving high accuracy requires fundamental increases in data collection proportional to the accuracy improvement demanded.

The Role of Class Complexity. The hypothesis class size appears as $\ln |\mathcal{H}|$, giving a logarithmic relationship. If the class size increases from $|\mathcal{H}| = 100$ to $|\mathcal{H}| = 10,000$ (a 100-fold increase), the required sample size increases by only $\frac{1}{\varepsilon} \ln 100 \approx \frac{4.6}{\varepsilon}$. This is remarkably favorable: even classes with millions of functions remain learnable with reasonable amounts of data.

However, this logarithmic advantage critically depends on finiteness. The bound becomes useless for infinite classes where $\ln|\mathcal{H}| = \infty$. For infinite classes—which include most real-world hypothesis classes like neural networks with continuous weights—we need fundamentally different analytical tools.

Let us illustrate these relationships with specific numbers. Suppose our hypothesis class has $|\mathcal{H}| = 10,000$ predictors, and we want accuracy $\varepsilon = 0.1$ and confidence $\delta = 0.05$ (95% success probability). The required sample size is:

$$n \geq \frac{1}{0.1} \ln \frac{10,000}{0.05} = 10 \cdot \ln(200,000) \approx 10 \cdot 12.2 = 122 \text{ examples}$$

Now suppose we improve accuracy to $\varepsilon = 0.02$:

$$n \geq \frac{1}{0.02} \ln(200,000) \approx 50 \cdot 12.2 = 610 \text{ examples}$$

A five-fold improvement in accuracy requires five times more data. By contrast, if we only increase confidence from 95% to 99% (reducing δ from 0.05 to 0.01) while keeping $\varepsilon = 0.1$:

$$n \geq \frac{1}{0.1} \ln \frac{10,000}{0.01} = 10 \cdot \ln(1,000,000) \approx 10 \cdot 13.8 = 138 \text{ examples}$$

Only 16 additional examples are needed for four times higher confidence, illustrating the logarithmic advantage.

Bad Datasets vs. Very Bad Datasets

It's worth clarifying the distinction between a bad dataset and a very bad dataset, as this affects how we interpret the bound. A bad dataset is one where at least one bad predictor achieves zero training error. However, ERM might not return that bad predictor — it might return another predictor with zero training error that is actually good. A very bad dataset is the worse case where ERM specifically returns a bad predictor.

The bound we have derived covers the probability of obtaining a bad dataset. Since ERM returns one of the zero-training-error predictors, being good or bad depends on which one it picks. In the worst case, it picks a bad one. However, in many practical scenarios, ERM will pick a good predictor among those with zero training error, especially if the dataset is large. The union bound is deliberately pessimistic, ensuring the bound holds in worst-case scenarios.

The Importance of the Finiteness Assumption

A critical observation is that this entire analysis relies on the hypothesis class being finite. The union bound, which aggregates failure probabilities across $|\mathcal{H}|$ predictors, would yield a meaningless bound (infinity) for infinite classes. This distinction—between finite and infinite classes—is profound:

- Every finite class, no matter how large, is learnable. The logarithmic dependence on $|\mathcal{H}|$ means even classes with trillions of functions are learnable with polynomial sample complexity.
- Infinite classes require different analysis. We cannot simply count the number of functions. Instead, we use measures of complexity like VC dimension, which capture the effective size of an infinite class.

This explains why the theoretical study of learning focuses so heavily on characterizing hypothesis class complexity. For finite classes, complexity is simply size. For infinite classes, we need more sophisticated notions.

PAC Learning: Formalizing Learnability

We now formalize the concepts developed in the previous sections. The framework we introduce is called **Probably Approximately Correct** (PAC) learning, and it provides a precise mathematical definition of what it means for a learning problem to be solvable. This definition will allow us to characterize exactly which hypothesis classes are learnable and to understand the role of sample complexity.

We begin with the simplest setting: when the realizability assumption holds and we use 0-1 loss for binary classification.

Definition 1 (PAC Learnability). *A hypothesis class $\mathcal{H}$ is* **PAC learnable** *if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ (the sample complexity function) and a learning algorithm $\mathcal{A}$ such that the following holds:*

For every accuracy parameter $\varepsilon \in (0,1)$, every confidence parameter $\delta \in (0,1)$, every distribution p_M over the domain $\mathcal{X}$, and every labeling function $f : \mathcal{X} \to \{0,1\}$, if the realizability assumption is satisfied (meaning there exists $h^* \in \mathcal{H}$ with $\mathcal{R}_{p_M,f}(h^*) = 0$), then when $\mathcal{A}$ is applied to a training set $\mathcal{T}$ of size $n \geq m_{\mathcal{H}}(\varepsilon, \delta)$ generated by sampling n independent examples from p_M and labeling them with f, the algorithm returns a predictor $h_{\mathcal{T}}$ such that

$$\mathcal{R}_{p_M,f}(h_{\mathcal{T}}) \leq \varepsilon$$

with probability at least $1 - \delta$ (over the randomness in sampling the training set).

This definition captures the essential meaning of learning. The accuracy parameter ε quantifies the "approximately correct" part: we don't demand perfect performance, only error at most ε. The confidence parameter δ quantifies the "probably" part: we don't need success for every possible training set, only for a fraction $1 - \delta$ of them. The sample complexity function $m_{\mathcal{H}}(\varepsilon, \delta)$ specifies how much data is necessary to achieve these goals, and it generally depends on the hypothesis class properties (for finite classes, it grows logarithmically with $|\mathcal{H}|$, as we've seen).

From the analysis in the previous section, we know that for finite hypothesis classes, ERM is a PAC learning algorithm with sample complexity:

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \left\lceil \frac{1}{\varepsilon} \ln \frac{|\mathcal{H}|}{\delta} \right\rceil$$

However, the definition allows for other algorithms $\mathcal{A}$ that might achieve better sample complexity than ERM. In principle, $m_{\mathcal{H}}(\varepsilon, \delta)$ could be smaller than the ERM bound above. In practice, for finite classes, ERM is optimal (up to constants), so achieving better dependence is not possible.

The Bayes Predictor

When we move beyond the realizability assumption, labels are generated stochastically. Let $p_C(t|\mathbf{x})$ denote the conditional probability of label t given features $\mathbf{x}$. In this probabilistic framework, there exists an optimal predictor that minimizes true risk:

$$h^*(\mathbf{x}) = \operatorname*{argmin}_{y \in \{0,1\}} \mathbb{E}_{t \sim p_C(\cdot|\mathbf{x})}[L(y, t)]$$

For binary classification with 0-1 loss, this optimal predictor has a particularly simple form. If we predict $y = 0$, the expected loss is $p_C(t = 1|\mathbf{x})$ (the probability we're wrong). If we predict $y = 1$, the expected loss is $p_C(t = 0|\mathbf{x}) = 1 - p_C(t = 1|\mathbf{x})$. We should predict whichever label incurs lower expected loss:

$$h_{\text{Bayes}}(\mathbf{x}) = \begin{cases} 1 & \text{if } p_C(t = 1|\mathbf{x}) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

This is the **Bayes predictor**, the best possible classifier for the problem. Its risk, denoted $\mathcal{R}^* = \mathcal{R}_p(h_{\text{Bayes}})$, represents the unavoidable error inherent in the stochasticity of label generation—no algorithm can do better.

The Bayes predictor is unattainable in practice because it requires knowledge of the true conditional distribution $p_C(t|\mathbf{x})$, which we never have. Moreover, a fundamental result called the *No Free Lunch theorem* (see later) states that without additional assumptions about the problem structure, no learning algorithm can guarantee finding a predictor as good as Bayes for every possible distribution. Learning is only possible because real problems have structure.

Agnostic PAC Learning

Since the Bayes predictor is unreachable and the realizability assumption rarely holds, we need a more pragmatic learning framework. In *agnostic* learning, we acknowledge that our hypothesis class might not contain any perfect predictor. Instead, we accept the best predictor available in our class, which has some non-zero error. We ask: can our learning algorithm find a predictor nearly as good as the best in the class?

Definition 2 (Agnostic PAC Learnability). *A hypothesis class $\mathcal{H}$ is* **agnostic PAC learnable** *if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm $\mathcal{A}$ such that:*

For every accuracy parameter $\varepsilon \in (0,1)$, every confidence parameter $\delta \in (0,1)$, and every distribution p over $\mathcal{X} \times \mathcal{Y}$, when $\mathcal{A}$ is applied to a training set $\mathcal{T}$ of size $n \geq m_{\mathcal{H}}(\varepsilon, \delta)$ generated by sampling n independent pairs from p, the algorithm returns a predictor $h_{\mathcal{T}}$ such that

$$\mathcal{R}_p(h_{\mathcal{T}}) \leq \min_{h \in \mathcal{H}} \mathcal{R}_p(h) + \varepsilon$$

with probability at least $1 - \delta$ *(over the randomness in sampling).*

This definition is more modest than PAC learnability. We don't compare ourselves to zero error or to the Bayes risk. Instead, we compare to the best error achievable within our hypothesis class, denoted $\mathcal{R}_p(h^*) = \min_{h \in \mathcal{H}} \mathcal{R}_p(h)$. This decomposition separates learning into two challenges: approximation error (how well the best in our class approximates the truth) and estimation error (how well we estimate the best in the class). Statistical learning theory addresses estimation error; approximation error is addressed by choosing good hypothesis classes.

Notice that agnostic PAC learnability generalizes PAC learnability: if the realizability assumption holds, the best predictor in the class has zero error, so the definition reduces to ordinary PAC learning.

Extending to General Loss Functions

The framework extends naturally beyond 0-1 loss to any bounded loss function.

Definition 3 (Agnostic PAC Learnability with General Loss)**.** *A hypothesis class* $\mathcal{H}$ *is* **agnostic PAC learnable** *with respect to a loss function* $L : \mathcal{Y} \times \mathcal{Y} \to [0, M]$ *if there exist a function* $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ *and a learning algorithm* $\mathcal{A}$ *such that:*

For every $\varepsilon, \delta \in (0,1)$ *and every distribution* p *over* $\mathcal{X} \times \mathcal{Y}$*, when* $\mathcal{A}$ *is applied to a training set* $\mathcal{T}$ *of size* $n \geq m_{\mathcal{H}}(\varepsilon, \delta)$*, it returns* $h_{\mathcal{T}}$ *satisfying*

$$\mathcal{R}_p(h_{\mathcal{T}}) = \mathbb{E}_{(\mathbf{x},t)\sim p}[L(h_{\mathcal{T}}(\mathbf{x}), t)] \leq \min_{h \in \mathcal{H}} \mathbb{E}_{(\mathbf{x},t)\sim p}[L(h(\mathbf{x}), t)] + \varepsilon$$

with probability at least $1 - \delta$*.*

This generalization allows application to regression, cost-sensitive classification, and many other learning problems beyond binary classification with 0-1 loss.

Empirical Risk vs True Risk: Representative Training Sets

For ERM to work, a critical condition must hold: empirical risk must be a good proxy for true risk. Specifically, the empirical risk of every predictor in $\mathcal{H}$ should closely approximate its true risk. This is formalized through the concept of a representative training set.

Definition 4 (ε-Representative Sample)**.** *A training set* $\mathcal{T}$ *is* ε*-representative with respect to hypothesis class* $\mathcal{H}$*, loss function* L*, and distribution* $p(\mathbf{x}, t)$ *if for every predictor* $h \in \mathcal{H}$*:*

$$\left|\overline{\mathcal{R}}_{\mathcal{T}}(h) - \mathcal{R}_p(h)\right| \leq \varepsilon$$

In other words, each predictor's training error differs from its true error by at most ε*.*

This definition ensures that the rankings of predictors are preserved: if one predictor is better than another on the population, it will likely be better on the training set as well (up to an ε margin).

Why does representativeness matter? Suppose a training set $\mathcal{T}$ is $\frac{\varepsilon}{2}$-representative. Then for the predictor $h_{\mathcal{T}}$ returned by ERM, we have:

By representativeness, for the optimal predictor $h^* = \arg\min_{h \in \mathcal{H}} \mathcal{R}_p(h)$:

$$\overline{\mathcal{R}}_{\mathcal{T}}(h^*) \leq \mathcal{R}_p(h^*) + \frac{\varepsilon}{2}$$

Since ERM minimizes empirical risk:

$$\overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) \leq \overline{\mathcal{R}}_{\mathcal{T}}(h^*) \leq \mathcal{R}_p(h^*) + \frac{\varepsilon}{2}$$

Finally, by representativeness again:

$$\mathcal{R}_p(h_{\mathcal{T}}) \leq \overline{\mathcal{R}}_{\mathcal{T}}(h_{\mathcal{T}}) + \frac{\varepsilon}{2} \leq \mathcal{R}_p(h^*) + \varepsilon$$

Thus, representativeness directly guarantees the agnostic PAC property! If we can show that a training set is representative with high probability, we've solved the learning problem.

The question now becomes: when can we guarantee that a random training set is representative? The answer is provided by **uniform convergence**.

Definition 5 (Uniform Convergence). *A hypothesis class $\mathcal{H}$ has the* **uniform convergence** *property with respect to a loss function L if there exists a function $m_{\mathcal{H}}^{UC} : (0,1)^2 \to \mathbb{N}$ such that:*

For every $\varepsilon, \delta \in (0,1)$ and every distribution $p(\mathbf{x}, t)$, if $\mathcal{T}$ is a dataset of $n \geq m_{\mathcal{H}}^{UC}(\varepsilon, \delta)$ independent pairs drawn from p, then with probability at least $1 - \delta$, the sample $\mathcal{T}$ is ε-representative.

The intuition is subtle: it's not enough that a single predictor converges to its true risk. We need *all* predictors in the class to simultaneously converge. This is harder than standard statistical convergence because we're making $|\mathcal{H}|$ statements at once.

For finite classes, the union bound directly provides uniform convergence. We can bound the probability that any predictor's empirical risk deviates from its true risk, sum over all $|\mathcal{H}|$ predictors (ensuring each has only $\frac{\delta}{|\mathcal{H}|}$ failure probability), and ensure the total failure probability is at most δ. This gives:

$$m_{\mathcal{H}}^{\mathrm{UC}}(\varepsilon, \delta) \leq \left\lceil \frac{1}{\varepsilon^2} \ln \frac{2|\mathcal{H}|}{\delta} \right\rceil$$

(This bound uses concentration inequalities like Chernoff or Hoeffding bounds, which provide tighter control than the union bound used for the realizability case.)

Summary

We've now established a chain of implications:

$$\text{Uniform Convergence} \implies \varepsilon\text{-Representativeness} \implies \text{Agnostic PAC Learnability}$$

More precisely: if $\mathcal{H}$ has uniform convergence property with function $m_{\mathcal{H}}^{\text{UC}}$, then $\mathcal{H}$ is agnostic PAC learnable with sample complexity:

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq m_{\mathcal{H}}^{\text{UC}}\left(\frac{\varepsilon}{2}, \delta\right)$$

and ERM is a successful agnostic PAC learning algorithm for $\mathcal{H}$.

This means that to prove agnostic PAC learnability, we only need to prove uniform convergence. And for finite classes, uniform convergence is automatic via the union bound.

The theoretical framework introduced above provides a complete characterization of when learning is possible:

For finite hypothesis classes, we have proven uniform convergence, which implies agnostic PAC learnability, which means ERM will find a predictor nearly as good as the best in the class. The sample complexity has a clear dependence on accuracy ($1/\varepsilon^2$ in the agnostic case, $1/\varepsilon$ in the realizable case), confidence ($\ln(1/\delta)$), and class size ($\ln|\mathcal{H}|$).

For infinite classes, this simple counting argument fails. We must use more sophisticated complexity measures like VC dimension or Rademacher complexity, which capture the effective size of the class. These tools extend the same principles to infinite classes, but with more subtle mathematics.

The key insight throughout is that learning is not magic: it's governed by simple mathematical laws. The amount of data required depends predictably on how complex our hypothesis class is, how accurate we want to be, and how confident we need to be. Understanding these relationships allows us to design learning algorithms effectively and to recognize when a learning problem is fundamentally solvable.

PAC learnability of finite classes

Let $\mathcal{H}$ be a finite hypothesis class, let $\mathcal{X} \times \mathcal{Y}$ be a domain, and let $l : \mathcal{X} \times \mathcal{Y} \mapsto [0, 1]$ be a loss function. Then, it is possible to prove that $\mathcal{H}$ enjoys the uniform convergence property with sample complexity

$$m_{\mathcal{H}}^{UC}(\varepsilon, \delta) \leq \left\lceil \frac{1}{2\varepsilon^2} \ln \frac{2|\mathcal{H}|}{\delta} \right\rceil$$

As a consequence, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq m_{\mathcal{H}}^{UC}\left(\frac{\varepsilon}{2}, \delta\right) \leq \left\lceil \frac{1}{\varepsilon^2} \ln \frac{2|\mathcal{H}|}{\delta} \right\rceil$$

That is, if the number of possible predictors is m, then by first sampling

$$\left\lceil \frac{1}{\varepsilon^2} \ln \frac{2|\mathcal{H}|}{\delta} \right\rceil = \left\lceil \frac{2(\ln m - \ln \delta + \ln 2)}{\varepsilon^2} \right\rceil$$

i.i.d. examples from $\mathcal{X} \times \mathcal{Y}$, each one under probability distribution $p(x, y)$, and next applying *ERM* on the resulting dataset, we obtain with probability $1 - \delta$ a predictor whose expected loss over pairs in $\mathcal{X} \times \mathcal{Y}$ is the one of best predictor among the m considered, plus at most ε.

Observe that the number of examples grows logarithmically wrt m while decreases (logarithmically) wrt to δ and (as a square root) wrt ε.

While this result only applies to finite hypothesis classes, there is a simple trick that allows us to get a very good estimate of the practical sample complexity of infinite hypothesis classes.

Consider a hypothesis class that is parameterized by d parameters. For example, let $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{-1, +1\}$, and the hypothesis class, $\mathcal{H}$, be all functions of the form $h_\theta(x) = \operatorname{sgn}(x - \theta)$. That is, each hypothesis is parameterized by one parameter, $\theta \in \mathbb{R}$, and the hypothesis outputs 1 for all instances larger than θ and outputs -1 for instances smaller than θ. This is a hypothesis class of an infinite size. However, if we are going to learn this hypothesis class in practice, using a computer, we will probably maintain real numbers using floating point representation, say, of 64 bits. It follows that in practice, our hypothesis class is parameterized by the set of scalars that can be represented using a 64 bits floating point number. There are at most 2^{64} such numbers; hence the actual size of our hypothesis class is at most 2^{64}. More generally, if our hypothesis class is parameterized by d numbers, in practice we learn a hypothesis class of size at most 2^{64d}. By the above considerations, we obtain that the sample complexity of such classes is bounded by

$$\frac{128d + 2\ln\frac{2}{\delta}}{\varepsilon^2} = \frac{128d + 2\ln 2 - 2\ln\delta}{\varepsilon^2}$$

This upper bound on the sample complexity has the deficiency of being dependent on the specific representation of real numbers used by our machine. In the following we will introduce a rigorous way to analyze the sample complexity of infinite size hypothesis classes. Nevertheless, the discretization trick can be used to get a rough estimate of the sample complexity in many practical situations.

Relevance of the inductive bias

Whenever we choose a hypothesis class $\mathcal{H}$, we actually make use of some prior knowledge about our data. We only choose the class because we believe (or hope) that it contains a good (i.e. low-risk) predictor for the task we are considering.

This begs the question whether such prior knowledge is a necessary condition for learning, or whether there can exist a "universal learner". Recall that a learning task is defined by an unknown probability distribution p over the set of all possible examples and labelings $\mathcal{X} \times \mathcal{Y}$, and given a training set of size n, we want to find a hypothesis $h \in \mathcal{H}$ that has a low risk with high probability. So a universal learner would correspond to an algorithm $\mathcal{A}^*$ and a sample complexity n such that $\mathcal{A}^*$ finds a low risk predictor $\mathcal{H}$ with high probability, whatever is the distribution p.

The fundamental question of whether a universal learner can exist is settled by the **No-Free-Lunch (NFL) Theorem**. This theorem asserts that for binary classification, no single learning algorithm can succeed on every possible task. More precisely, for any given algorithm, there exists a distribution on which it performs poorly, even though the task is simple (realizable). This negative result formalizes the necessity of inductive bias—prior knowledge that guides the choice of hypothesis class.

Theorem 1 (No-Free-Lunch)**.** *Let $\mathcal{X}$ be a domain, and let $\mathcal{A}$ be any learning algorithm for binary classification with* 0-1 *loss. For any sample size $n < |\mathcal{X}|/2$, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$ such that:*

1. *Realizability: There exists a function $h^* : \mathcal{X} \to \{0, 1\}$ with zero risk, i.e., $L_{\mathcal{D}}(h^*) = 0$.*
2. *High error with high probability: When $\mathcal{A}$ is trained on a sample $S \sim \mathcal{D}^n$, the resulting hypothesis h_S satisfies*

$$\mathbb{P}_{S\sim\mathcal{D}^n}\left[L_{\mathcal{D}}(h_S) \geq \frac{1}{8}\right] \geq \frac{1}{7}.$$

The theorem highlights a stark limitation: for every learner $\mathcal{A}$, one can construct an adversarial distribution $\mathcal{D}$ that makes $\mathcal{A}$ fail. Importantly, $\mathcal{D}$ is realizable (there exists a perfect predictor), yet $\mathcal{A}$—regardless of its sophistication—produces a hypothesis with error at least $1/8$ with probability at least $1/7$. Note that the constants $1/8$ and $1/7$ are not sharp; they can be improved, but the qualitative message remains: no algorithm is universally successful.

Implications for PAC Learnability

The NFL theorem directly implies that the class of *all* binary-valued functions on an infinite domain is not PAC learnable. This is a crucial fact that underscores the need for restricting the hypothesis class.

Theorem 2. *Let $\mathcal{X}$ be an infinite set, and let $\mathcal{F} = \{f : \mathcal{X} \to \{0, 1\}\}$ be the set of all possible binary classifiers on $\mathcal{X}$. Then $\mathcal{F}$ is not PAC learnable.*

Dimostrazione. Assume, for contradiction, that $\mathcal{F}$ is PAC learnable. Then there exists an algorithm $\mathcal{A}_{\mathcal{F}}$ and a function $m_{\mathcal{F}} : (0, 1)^2 \to \mathbb{N}$ such that for every $\varepsilon, \delta \in (0, 1)$ and every distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$ for which $\min_{h\in\mathcal{F}} L_{\mathcal{D}}(h) = 0$ (i.e., realizability holds), we have

$$\mathbb{P}_{S\sim\mathcal{D}^m}\left[L_{\mathcal{D}}(\mathcal{A}_{\mathcal{F}}(S)) \leq \varepsilon\right] \geq 1 - \delta,$$

provided $m \geq m_{\mathcal{F}}(\varepsilon, \delta)$.

Now choose $\varepsilon = 1/16$ and $\delta = 1/8$ (note $\varepsilon < 1/8$ and $\delta < 1/7$). Let $n = m_{\mathcal{F}}(\varepsilon, \delta)$. Since $\mathcal{X}$ is infinite, we have $n < |\mathcal{X}|/2$. By the NFL theorem applied to $\mathcal{A}_{\mathcal{F}}$ and n, there exists a distribution $\mathcal{D}$ such that:

- There exists $h^* \in \mathcal{F}$ with $L_{\mathcal{D}}(h^*) = 0$ (so realizability holds).
- $\mathbb{P}_{S\sim\mathcal{D}^n}\left[L_{\mathcal{D}}(\mathcal{A}_{\mathcal{F}}(S)) \geq 1/8\right] \geq 1/7 > \delta$.

But this contradicts the PAC guarantee, which requires that the probability of error exceeding $\varepsilon = 1/16$ is at most $\delta = 1/8$. Hence, $\mathcal{F}$ cannot be PAC learnable. □

The corollary demonstrates that without any inductive bias—that is, by considering the class of all possible functions—learning is impossible. The NFL theorem constructs a distribution that exploits the algorithm's inability to generalize beyond the training set when the hypothesis space is too large.

The Role of Inductive Bias

The failure of universal learning forces us to incorporate prior knowledge by restricting the hypothesis class $\mathcal{H}$ to a proper subset of $\mathcal{F}$. This restriction is the inductive bias: the assumption that a good predictor belongs to $\mathcal{H}$. The bias-variance decomposition quantifies the trade-off involved in choosing $\mathcal{H}$:

- A small $\mathcal{H}$ may have large *approximation error* (bias) but small *estimation error* (variance).
- A large $\mathcal{H}$ may reduce approximation error but increase estimation error, leading to overfitting.

The NFL theorem shows that if we choose $\mathcal{H} = \mathcal{F}$, the estimation error becomes uncontrollable. Therefore, successful learning requires a carefully chosen $\mathcal{H}$ that balances bias and variance based on prior knowledge about the task. This is the essence of model selection: picking a hypothesis class (or a family of classes) that is sufficiently expressive to approximate the target function yet sufficiently constrained to allow generalization from finite data.

In summary, the No-Free-Lunch theorem establishes that learning is impossible without assumptions. Inductive bias, embodied by the choice of hypothesis class, is not a mere convenience but a fundamental necessity for generalization.

From the No Free Lunch Theorem we can conclude that choosing a suitable hypothesis class is crucial for learning a given concept. This way we restrict ourselves to a subset of all possible functions from $\mathcal{X}$ $\{0, 1\}$, which helps us avoiding unfavourable distributions and might allow us to find a low-error hypothesis with high probability.

On the other hand we might exclude the best hypotheses from our set of candidates, as it might not be a member of our hypothesis class. So we might find a good approximation for the best hypothesis in our class, but this best hypothesis in the class might be a bad approximation for the true target predictor. This dilemma is often referred to as the Bias-Complexity Tradeoff .

If we choose a hypothesis class $\mathcal{H}$ and get a training set $\mathcal{T}$ as input, ERM returns a predictor $h_{\mathcal{T}}$ minimizing such that the empirical risk $\mathcal{R}_p(h)$ is minimal.

Now we can decompose this risk as follows:

$$\mathcal{R}_p(h_{\mathcal{T}}) - \mathcal{R}_p(h_{\text{Bayes}}) = \underbrace{(\mathcal{R}_p(h_{\mathcal{T}}) - \mathcal{R}_p(h^*)}_{\text{estimation error}} + \underbrace{(\mathcal{R}_p(h^*) - \mathcal{R}_p(h_{\text{Bayes}}))}_{\text{approximation error}} = \varepsilon_V + \varepsilon_B$$

where h^* is the best predictor in $\mathcal{H}$ (that is such that $\mathcal{R}_p(h^*) = \min_{h \in \mathcal{H}} \mathcal{R}_p(h^*)$, while h_{Bayes} is the absolute best predictor for the task (that is such that $\mathcal{R}_p(h_{\text{Bayes}}) = \min_{h \in \mathcal{F}} \mathcal{R}_p(h^*)$, where $\mathcal{F}$ is the set of all possible predictors).

More in detail:

- ϵ_B is the minimum risk achievable by any $h \in \mathcal{H}$: this is only determined by the inductive bias, and independent from the training set. It is a property of the class of hypotheses considered with respect to the prediction task. This is called **bias**
- ϵ_V is the difference between the above minimum risk in $\mathcal{H}$ and the risk associated to the best predictor in $\mathcal{H}$ with respect to the training set: it is related to the fact that empirical risk minimization only provides an estimate of the best predictor achievable for the given inductive bias. It is a measure of how well the

predictor computed from a particular training set approximates the best possible one. Its expectation with respect to all possible training sets is a measure of how much a predictor derived from a random training set may result in poorer performances with respect to the best possible one. This is called **variance**

The Bias–Variance Trade-off in Model Selection

The choice of hypothesis class $\mathcal{H}$ involves a fundamental tension known as the **bias–variance trade-off**. This trade-off arises from the above decomposition of the excess risk, where the approximation error (bias) ϵ_B measures how well the optimal predictor in $\mathcal{H}$ can approximate the Bayes optimal predictor $h^*_{\mathcal{D}}$. This error depends solely on the *expressive power* of $\mathcal{H}$ relative to the true data-generating distribution $\mathcal{D}$. The estimation error (variance) ϵ_V measures the additional error incurred because we only observe a finite sample S rather than the entire distribution.

The complexity of $\mathcal{H}$—often quantified by measures like VC dimension or Rademacher complexity—directly influences this trade-off:

1. High Bias, Low Variance (Underfitting): When $\mathcal{H}$ is too simple (e.g., linear models for a nonlinear relationship), the approximation error is large because even the best predictor in $\mathcal{H}$ cannot capture the underlying patterns. However, the estimation error is small: different training sets S yield similar predictors h_S, all with high but consistent error. This situation is called *underfitting*. The model fails to capture the signal in the data, leading to poor performance on both training and test sets.
2. Low Bias, High Variance (Overfitting): When $\mathcal{H}$ is very complex (e.g., high-degree polynomials or deep neural networks), the approximation error can be very small—$\mathcal{H}$ may even contain the Bayes optimal predictor. However, the estimation error becomes large because the algorithm has too much flexibility. Different training sets can produce wildly different predictors h_S, some of which fit the training data perfectly but generalize poorly. This is *overfitting*: the model captures noise in the training set as if it were signal, leading to low training error but high test error.

The Trade-off in Practice

In practice, one rarely fixes a single hypothesis class $\mathcal{H}$. Instead, we consider a nested family of classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots \subset \mathcal{H}_k$, where complexity increases with k. For example:

- Polynomial regression with increasing degree.
- Decision trees with increasing depth.
- Neural networks with increasing width or depth.

As we increase k, the approximation error (bias) decreases monotonically, while the estimation error (variance) increases. In general, if we choose a very large space $\mathcal{H}$, then the approximation term will become small (the Bayes classifier might even be contained in $\mathcal{H}$ or can be approximated closely by some element in it). The estimation error, however, will be rather large in this case: the space $\mathcal{H}$ will contain complex functions which will lead to overfitting. The opposite effect will happen if the function class $\mathcal{H}$ is very small.

The total risk (sum of both errors) typically exhibits a U-shaped curve as a function of complexity (see Figure 2). The optimal complexity is where this total risk is minimized.

The bias–variance trade-off provides a conceptual framework for understanding generalization. It emphasizes that learning is not merely about minimizing training error, but about finding the right balance between expressivity and simplicity.

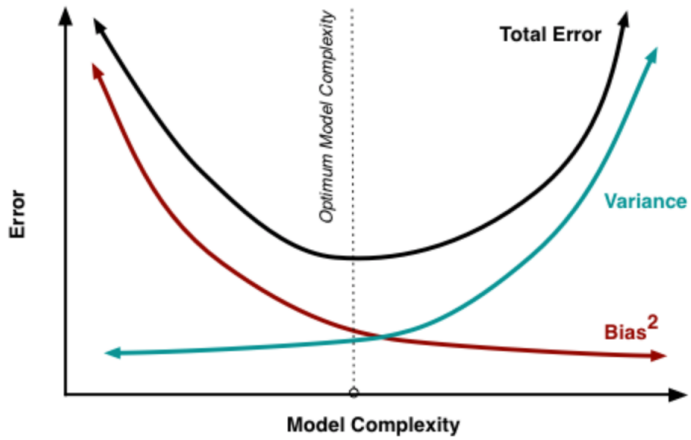

Figura 2. The bias–variance trade-off: as model complexity increases, bias decreases but variance increases. The total risk (solid curve) has a minimum at the optimal complexity.

See Figure 2 for an illustration. If we choose a very large space $\mathcal{H}$, then the approximation term will become small (the Bayes classifier might even be contained in $\mathcal{H}$ or can be approximated closely by some element in it). The estimation error, however, will be rather large in this case: the space $\mathcal{H}$ will contain complex functions which will lead to overfitting. The opposite effect will happen if the function class $\mathcal{H}$ is very small.

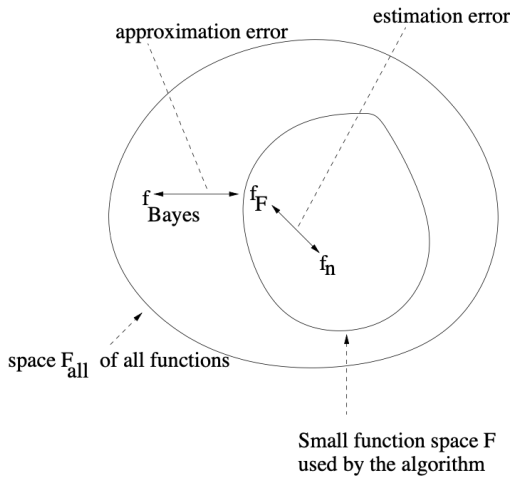

Figura 3. Illustration of estimation and approximation error.

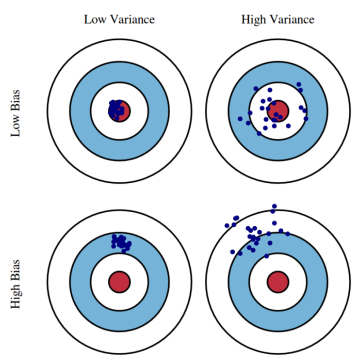

Figura 4. Graphical representation of bias and variance

Learning theory studies how rich we can make the class $\mathcal{H}$ while still maintaining a reasonable estimation error. In many cases, empirical research focuses on designing good classes of predictors for a given domain. Here, "good" means classes for which the approximation error should not be excessively high. The idea is that, even if we are not experts and do not know how to construct the optimal predictor, we still have some preliminary knowledge of the specific problem, which allows us to design classes of predictors for which both the approximation error and the estimation error are not too high.

In practice, one typically has a set of predictors with associated sets of hyper-parameters: a type of predictor and an assignment of values to the corresponding hyper-parameters define a class of parametric predictors, from which a specific predictor is selected through a learning algorithm such as ERM.

The identification of this class, by evaluating and selecting a type of predictor and the values of the corresponding hyper-parameters, is called **model selection**.

The Vapnik-Červonenkis Dimension

The *Vapnik-Červonenkis dimension* (VC-dimension) is a measure of the complexity of a given hypothesis class. The measure is introduced to characterize infinite hypothesis classes in terms of their learnability.

We know that finite hypothesis classes are PAC-learnable and that the sample complexity depends on the size of the class. However, there also exist infinite hypothesis classes that are PAC-learnable, such as the class of threshold functions on real numbers

$$\mathcal{H}_\theta = \{\mathbb{1}[x < \theta]; \theta \in \mathbb{R}\}.$$

This hypothesis class is PAC-learnable with a sample complexity of $m_{\mathcal{H}}(\varepsilon, \delta) \leq \lceil \frac{1}{\varepsilon} \log \frac{2}{\delta} \rceil$.

Thus, finiteness is sufficient but not necessary for learnability. To define a more general and useful measure of complexity, we first need a few other definitions.

Given a subset $C = \{c_1, ..., c_m\} \subset \mathcal{X}$ of $\mathcal{X}$, we define the *restriction* $\mathcal{H}_C$ of $\mathcal{H}$ to C as the set of functions $f : C \mapsto \{0, 1\}$ that can be derived from predictors in $\mathcal{H}$ (i.e., such that for each $f \in C$ there exists a predictor $h \in \mathcal{H}$ for which $f(c_i) = h(c_i), i = 1, \dots, m$). If we describe each function from C to $\{0, 1\}$ as a vector in $\{0, 1\}^{|C|}$, we can formally write it as

$$\mathcal{H}_C = \{(h(c_1), ...h(c_m)) : h \in \mathcal{H}\}.$$

If $\mathcal{H}_C$ is the set of all functions from C to $\{0, 1\}$ (and thus $|\mathcal{H}_C| = 2^{|C|}$), this means that for every binary labeling of the elements of C (and thus for every separation of the elements into two distinct classes, or even for every possible binary classification task on C), there exists a predictor in $\mathcal{H}$ that separates the two classes, in the sense that it correctly predicts the target values of each element c_i. In this case, we say that $\mathcal{H}$ *shatters* C.

For example, consider the class $\mathcal{H}_\theta$ of threshold functions introduced above. Consider a set $C = \{c_1\}$: then, if we set $\theta = c_1 + 1$, we have $h_\theta(c_1) = 1$, and if we take $\theta = c_1 - 1$, we have $h_\theta(c_1) = 0$. Therefore, $\mathcal{H}_C$ is the set of all functions from C to $\{0, 1\}$, and $\mathcal{H}$ shatters C. If we now consider a set $C = \{c_1, c_2\}$, where $c_1 \leq c_2$, no $h \in \mathcal{H}$ can represent the labeling $c_1 = 0, c_2 = 1$, because any threshold that assigns label 0 to c_1 must also assign label 0 to c_2. Therefore, not all functions from C to $\{0, 1\}$ are included in $\mathcal{H}_C$, so C is not shattered by $\mathcal{H}$.

We now define the VC-Dimension $\text{VCdim}(\mathcal{H})$ of a hypothesis class $\mathcal{H}$ as the size of the largest subset $C \subset \mathcal{X}$ that is shattered by $\mathcal{H}$.

The motivation behind this definition is the following. From the No-Free-Lunch theorem, we know that the set of all functions from a domain to $\{0, 1\}$ is not PAC-learnable. However, the proof of this statement is based on the assumption that we are considering all possible functions: it is reasonable to assume that introducing limitations on the hypothesis class might bring advantages.

To illustrate the concept of VC-Dimension, let us consider a few examples:

- Let $\mathcal{H}^{\text{thr}}$ be the class of threshold functions on $\mathbb{R}$ (tahat is functions h_θ such that $h_\theta(x) = 1$ if $x \geq \theta$ and $h_\theta(x) = 0$ otherwise). It is easy to see that for an arbitrary set $C = c_1$ of size 1, $\mathcal{H}_{\text{thr}}$ shatters C (just set $\theta = c_1$ if c_1 has label 1 and $\theta = c_1 + 1$ if the label is 0); therefore, $\text{VCdim}(\mathcal{H}^{\text{thr}}) \geq 1$. It is also possible to

check that there exist sets $C = \{c_1, c_2\}$ of size 2 that $\mathcal{H}$ does not shatter (consider the case $c_1 > c_2$ with labeling $c_1 = 1, c_2 = 0$). Therefore, we conclude that $\text{VCdim}(\mathcal{H}^{\text{thr}}) = 1$.

- Let $\mathcal{H}^{\text{rect}}$ be the set of all axis-aligned rectangles in the Euclidean plane. If we want to prove that a certain number d is indeed the VC-Dimension of a hypothesis class $\mathcal{H}^{\text{rect}}$, we need to prove two things: $\text{VCdim}(\mathcal{H}^{\text{rect}}) \geq d$ and $\text{VCdim}(\mathcal{H}^{\text{rect}}) < d + 1$.
  In this case, $\text{VCdim}(\mathcal{H}^{\text{rect}}) = 4$. For the first inequality, we simply need to find a set of 4 points shattered by axis-aligned rectangles. Consider 4 equidistant points on a circle.

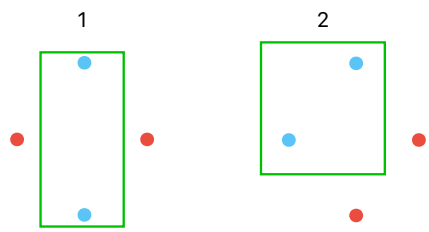

Figura 5. The two possible cases of shattering a set of 4 elements using axis-aligned rectangles.

It is easy to see (Figure 5) that any partition of the 4 points into positive (green) and negative (red) can be shattered by an axis-aligned rectangle.
To prove that $\text{VCdim}(\mathcal{H}^{\text{rect}}) < 5$, we need to show that *no* set of size 5 is shattered. To this end, consider an arbitrary set of 5 points (Figure 6) and their bounding box (indicated by the dashed green rectangle). There must be at least one point inside the bounding box (or on one of its sides). Now, if we label the other 4 points as positive and the fifth as negative, any axis-aligned rectangle that contains all the positive points must also contain the negative point. Therefore, no set of size 5 is shattered by $\mathcal{H}^{\text{rect}}$.

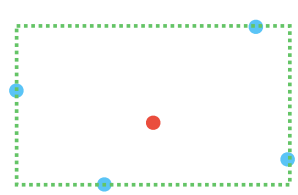

Figura 6. The impossibility of shattering a set of 5 elements using axis-aligned rectangles.

Let $\mathcal{H}^{\text{int}}$ be the class of intervals on $\mathbb{R}$, precisely $\mathcal{H}^{\text{int}} = \{h_{a,b} : a, b \in \mathbb{R}, a < b\}$, where $h_{a,b} : \mathbb{R} \to \{0, 1\}$ is a function such that $h_{a,b}(x) = \mathbb{1}[a < x < b]$. Consider the set $C = \{1, 2\}$. In this case, $\mathcal{H}^{\text{int}}$ shatters C (Fig. 7), and therefore $\text{VCdim}(\mathcal{H}^{\text{int}}) \geq 2$. Now, take an arbitrary set $C = \{c_1, c_2, c_3\}$ and assume without

loss of generality that $c_1 \leq c_2 \leq c_3$. In this case, the labeling $(1, 0, 1)$ cannot be obtained from an interval, so $\mathcal{H}^{\text{int}}$ does not shatter C. Therefore, we conclude that $\text{VCdim}(\mathcal{H}^{\text{int}}) = 2$.

Figura 7. Shattering of a 2-element set using intervals.

- Let $\mathcal{H}^{\text{fin}}$ be a finite class. As observed, in order to shatter a set C we need $2^{|C|}$ predictors (since $2^{|C|}$ is the number of different labelings). Also, observe that $|\mathcal{H}_C^{\text{fin}}| \leq |\mathcal{H}^{\text{fin}}|$. As a consequence, C cannot be shattered by $\mathcal{H}^{\text{fin}}$ if $|\mathcal{H}^{\text{fin}}| < 2^{|C|}$, since this implies $|\mathcal{H}_C^{\text{fin}}| < 2^{|C|}$. This results into $\text{VCdim}(\mathcal{H}^{\text{fin}}) \leq \log_2|\mathcal{H}|$. The PAC learnability of finite classes then derives from the more general property PAC learnability of classes with finite VC-dimension. However, note that the VC-dimension of a finite class $\mathcal{H}^{\text{fin}}$ can be significantly smaller than $\log_2(|\mathcal{H}^{\text{fin}}|)$. For example, let $\mathcal{X} = \{1, \dots, k\}$ for some integer k, and consider the class of threshold functions on $\mathcal{H}$. Then, $|\mathcal{H}| = k$ but $\text{VCdim}(\mathcal{H}) = 1$. Since k can be arbitrarily large, the difference between $\log_2(|\mathcal{H}|)$ and $\text{VCdim}(\mathcal{H})$ can be arbitrarily large.

The main reason for interest in studying the VC-dimension is the existence of the so-called **Fundamental Theorem of Statistical Learning**, which provides a direct connection between the VC-dimension and the PAC learnability of a hypothesis class.

Theorem 3 (Fundamental Theorem of Statistical Learning). *Let $\mathcal{H}$ be a class of hypotheses $h : \mathcal{X} \to \{0, 1\}$ for binary classification, and let the $0 - 1$ loss be the considered cost function. Then, the following statements are equivalent:*

1. *$\mathcal{H}$ has a finite VC-dimension.*
2. *$\mathcal{H}$ is agnostic PAC-learnable, and there exist constants $c_1 < c_2$ such that its sample complexity $m_{\mathcal{H}}(\varepsilon, \delta)$ is upper and lower bounded as*
$$\frac{c_1}{\varepsilon^2}\left(d + \ln\frac{1}{\delta}\right) \leq m_{\mathcal{H}}(\varepsilon, \delta) \leq \frac{c_2}{\varepsilon^2}\left(d + \ln\frac{1}{\delta}\right)$$
Moreover, this property holds also when ERM is applied (that is, it is a successful agnostic PAC-learning algorithm for $\mathcal{H}$).
3. *$\mathcal{H}$ is PAC-learnable, and its sample complexity $m_{\mathcal{H}}(\varepsilon, \delta)$ is upper and lower bounded as*
$$\frac{c_1}{\varepsilon}\left(d + \ln\frac{1}{\delta}\right) \leq m_{\mathcal{H}}(\varepsilon, \delta) \leq \frac{c_2}{\varepsilon}\left(d + \ln\frac{1}{\delta}\right)$$
Moreover, this property holds also when ERM is applied (that is, it is a successful PAC-learner for $\mathcal{H}$).